PowerFlex 525 AC Drive
Fault Code Reference Manual
Rockwell Automation / Allen-Bradley

SECTION 1: FAULT CODES

F004 UnderVoltage
Description: DC bus voltage fell below minimum level.
Possible Causes:
- Low incoming line voltage
- Loose input power connections
- Power line disturbance or momentary power loss

Corrective Actions:
1. Check incoming line voltage at the drive input terminals.
   Verify voltage is within drive rating (480VAC +/-10%).
2. Inspect input power connections for loose terminals or corrosion.
3. Check for other loads on the same power circuit causing voltage sag.
4. Verify DC bus voltage using a multimeter on the DC+ and DC- terminals.
5. If voltage dips are momentary, consider adding a line reactor or UPS.

F005 OverVoltage
Description: DC bus voltage exceeded maximum level.
Possible Causes:
- High incoming line voltage
- Regenerative energy from decelerating load
- Deceleration rate set too fast

Corrective Actions:
1. Check incoming line voltage - must not exceed drive maximum rating.
2. Increase deceleration time (Parameter A440).
3. Install a dynamic braking resistor if load is regenerative.
4. Verify DC bus voltage is within normal operating range (650-700VDC for 480V
).

SECTION 2: SAFETY WARNINGS

WARNING: LOCKOUT/TAGOUT REQUIRED
Before performing any maintenance on this drive:
- Disconnect and lockout all power sources per LOTO procedures.
- Wait 5 minutes after power removal before opening enclosure.
- Verify zero voltage with a calibrated meter before touching internal parts.
- Follow site lockout/tagout policy and OSHA 1910.147 requirements.

SECTION 3: MAINTENANCE INTERVALS

Monthly:
- Inspect drive cooling vents for blockage
- Check ambient temperature (must be below 50 deg C)
- Verify fan operation

Annual:
- Clean heat sink fins with compressed air (drive de-energized, LOTO applied)
- Inspect power connections for torque and corrosion
- Test ESTOP function

SECTION 4: SPECIFICATIONS

Drive model: 25B-D010N104
Input: 480VAC 3-Phase 50/60 Hz
Output: 0-480VAC 3-Phase
HP Rating: 5 HP (3.7 kW)
DC Bus Nominal: 675 VDC
Overload: 150% for 60 seconds